

WHAT THIS COURSE TEACHES

The SEL Real Time Automation Controller (RTAC) is the platform that ties everything else together. It is a protocol gateway, a data concentrator, a logic engine, and an IEC 61850 interface — all in one ruggedized DIN-rail device. In power utilities and critical infrastructure, the RTAC sits between the field device level and the SCADA master, translating protocols, filtering data, and implementing local automation that must survive communication failures.

Engineers who understand the RTAC can design substation automation architectures, configure protocol mappings between DNP3, Modbus, and IEC 61850, and implement local control logic using ACSELERATOR Architect. This course covers the full stack from hardware commissioning through advanced protocol gateway configuration.

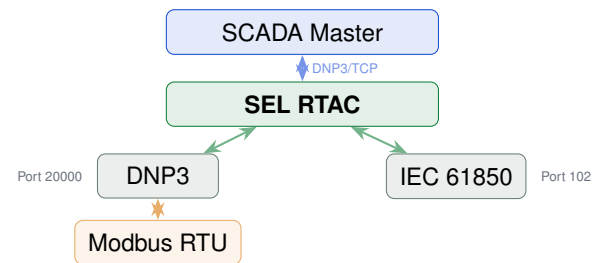
CHAPTER SEQUENCE

1. **RTAC Architecture** — Hardware variants (SEL-3505/3530/3555), I/O modules, communication ports, power requirements, hardware commissioning.
2. **ACSELERATOR Architect** — Software environment, project structure, database tags, tag aliasing, revision control.
3. **DNP3 Configuration** — DNP3 master and outstation roles, point mapping, event handling, unsolicited reporting configuration.
4. **Modbus Gateway** — Modbus TCP server configuration, register mapping from DNP3 points, coil and holding register layout.
5. **IEC 61850 Integration** — GOOSE messaging, Sampled Values, MMS client/server, logical node mapping to RTAC tags.
6. **Local Logic** — ACSELERATOR logic engine: IEC 61131-3 programs, event-driven automation, failover sequences.
7. **Security & Access Control** — Role-based access, RBAC configuration, serial port security, firmware update procedures.
8. **Field Scenarios** — Substation automation case studies: transformer bay control, feeder protection, load shedding automation.

THE STORY

Substation automation is the application domain where every protocol from this curriculum converges. A modern substation communicates via DNP3 to the SCADA master, via IEC 61850 GOOSE between protection relays, via Modbus to legacy meters, and via serial to older RTUs. The RTAC is the device that manages all of that simultaneously.

SEL dominates the power utility automation market in North America. Understanding the RTAC is not optional for engineers working in substations, distribution automation, or grid modernization. This course provides the configuration depth that the SEL documentation provides in six separate manuals — integrated into a learning sequence that connects each feature to the underlying protocol concepts taught in the prior courses.



Certification Relevance

- **SEL Certified Engineer** — SEL offers application engineer certification via their training center; RTAC configuration is a core track.
- **GICSP (GIAC)** — Substation automation and IEC 61850 security appear in ICS security certification exams.
- **ISA CCST** — Protocol gateway configuration and field device integration are tested topics.

Next Steps (Certs)

1. Complete DNP3 + RTAC guides together for full stack coverage.
2. Download ACSELERATOR Architect (free from selinc.com) and configure a lab project.
3. Attend SEL application engineering training (Pullman, WA or regional).
4. Pursue GICSP for the ICS security credential that pairs with RTAC expertise.

8 Chapters

4 Protocol

160+ Questions

